

**Resolving Reservoir Connectivity Uncertainty in an Untested Fault Block: Mudeni E6200
Case Study, Onshore Niger Delta**

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ABSTRACT

Located in the central part of the Niger Delta, the Mudeni Field is segmented into two blocks; the main block 'A' and a second block 'B' separated by a minor fault, the tip of which does not seem to close at the adjacent fault based on seismic interpretation. The presence of a geologic fault would restrict lateral migration of hydrocarbons across the fault boundaries in situations where the boundaries are sealing and hence varying fluid contacts would be seen across the different blocks. The implication of compartmentalization for field development strategy is that each block must be targeted separately by development wells. Mudeni main block A has been tested and proven with significant production while block B is currently carried as an appraisal opportunity. The uncertainty around the fault seal potential had to be adequately addressed as substantial volumes close to the ultimate recovery had been produced over the years from the main block. A non-sealing fault would imply that significant volumes may have migrated across from block B into the zone of production in block A. A detailed mapping of the E6200 seismic event on a 2014 Pre-stack Depth Migration (PSDM) data provided the basis for guided seismic amplitude extraction. Using depth converted time grids, amplitude versus depth profiles were generated and analyzed. Hydrocarbon–water contacts were estimated based on observed amplitude profiles. This technique was validated using actual logged contact information observed from wells in block A. Contact prediction technique validation was successful as it confirmed the logged hydrocarbon water contact in block A at a depth of 8656 ft tvdss as measured from existing wells. Predicted hydrocarbon water contact in block B was estimated to be at a depth of 8723 ft tvdss which is 67 ft deeper than the contact at block A. The study results suggest that Block A and Block B are compartmentalized having different fluid contacts.

Key words: *Fault, Contact, Uncertainty, Compartmentalization, Seismic, Amplitude*

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INTRODUCTION

The onshore Niger Delta hydrocarbon reservoirs are essentially comprised of clastic sand deposits sandwiched between relatively extensive shale beds or mud rocks with varying thicknesses depending on the predominant mechanism of sedimentation prevalent at the time of deposition. The extensive shale beds or mud rocks are impermeable to flow of fluids thus forming effective seals that prevent further migration of hydrocarbons in the upward direction within the reservoirs. Reservoir sands occur in varying thicknesses and areal extents. The reservoir thicknesses are essentially controlled by the volume of sediment available for deposition and the accommodation space available at the given time. The areal extent of the reservoir sands could be confined within channel margins or fragmented into reservoir blocks by laterally extensive faults which are syn-depositional geologic features. These faults, depending on the scale and other properties are often

good barriers to lateral fluid migration. There have been instances where these faults have been known to be non-sealing, thereby enabling connectivity of fluids across the boundaries.

GEOLOGIC SETTING

The Mudeni Field is located in the Central Swamp Depobelt of the Niger Delta (Fig. 1). The field is part of a macrostructure, bounded to the north by a WSW-ENE oriented, south-dipping fault system. The macrostructure is an E-W elongated rollover anticline with four-way dip closure and a collapsed crest (Fig. 2). The reservoir depositional facies vary from amalgamated stacked channel complexes which are vertically more homogeneous to lower net-to-gross shoreface complexes with some vertical baffling. The reservoir intervals are laterally continuous over tens of kilometer based on estimates from well log and seismic correlations. The E6200 structure is limited to the north by a roughly East-West trending major boundary listric growth fault and a Northeast–Southwest minor synthetic fault to the west. The bounding fault to the North together with dip closure to the South, Southwest and East provide the structural trap. The northerly upthrown block characterized mainly by restricted aquifer was tested and found to be wet by Mudeni well-20. The hydrocarbon bearing downthrown block of the Mudeni Field is segmented into two reservoir blocks - the main block ‘A’ and a second block ‘B’ by what seemed to be a minor fault (Fig. 3). From seismic evaluation, the throw of the fault appears to be insignificant towards the tip. This gives rise to uncertainty if the fault is a competent seal or not, at the crestal area of the structure.

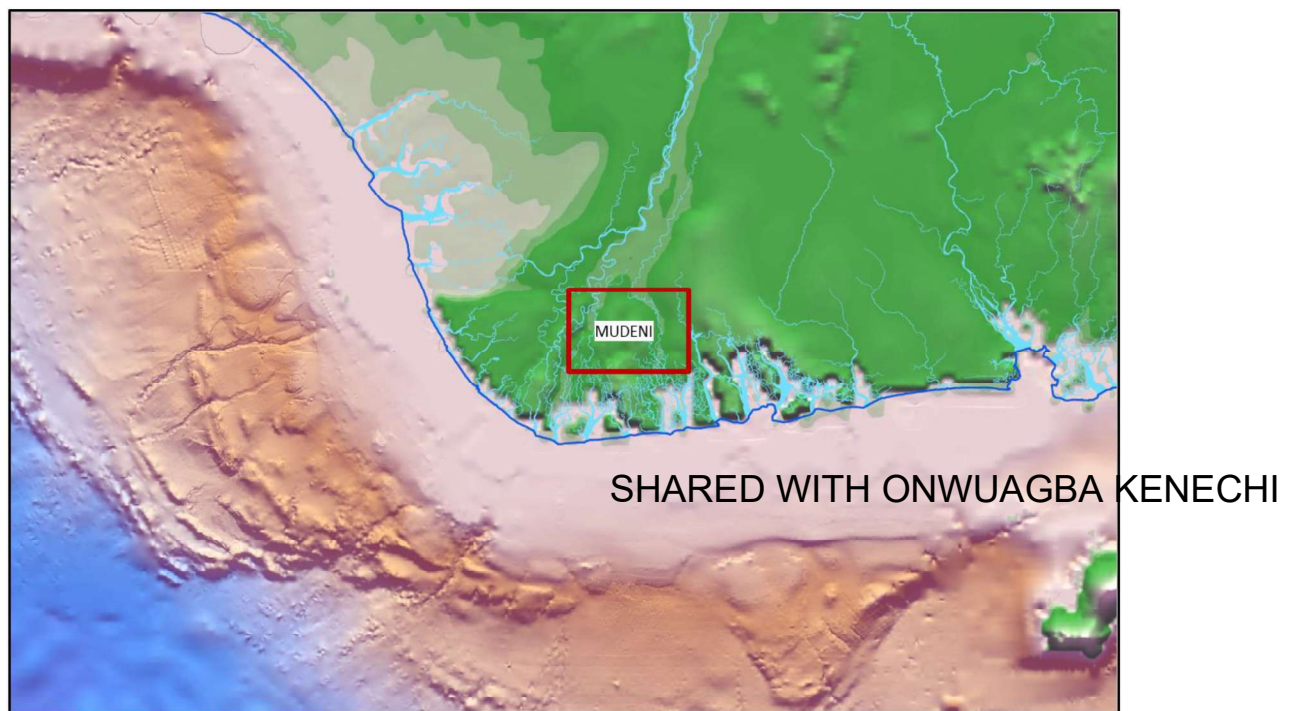


Figure 1: Location of the Study Area

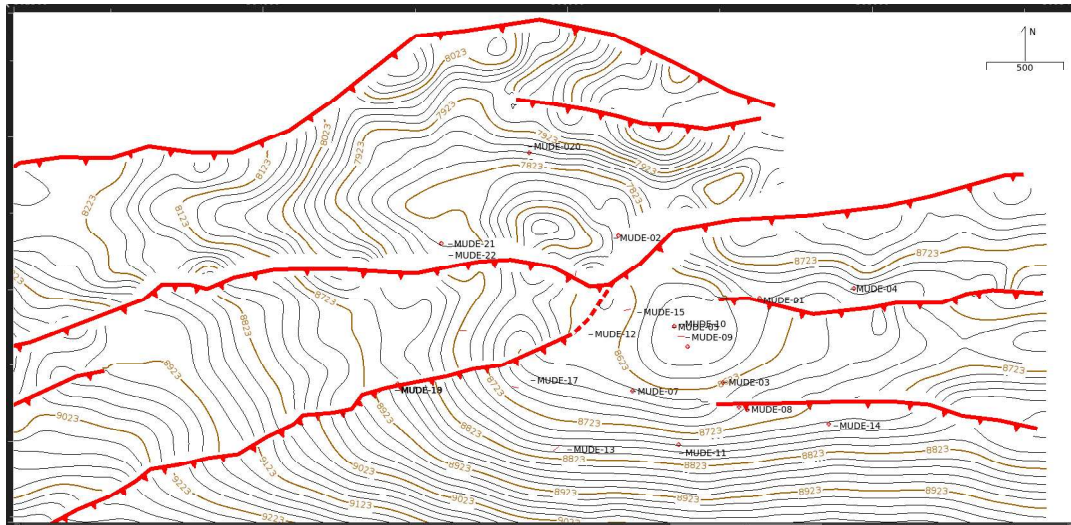


Figure 2: Mudeni Macro Structural Framework

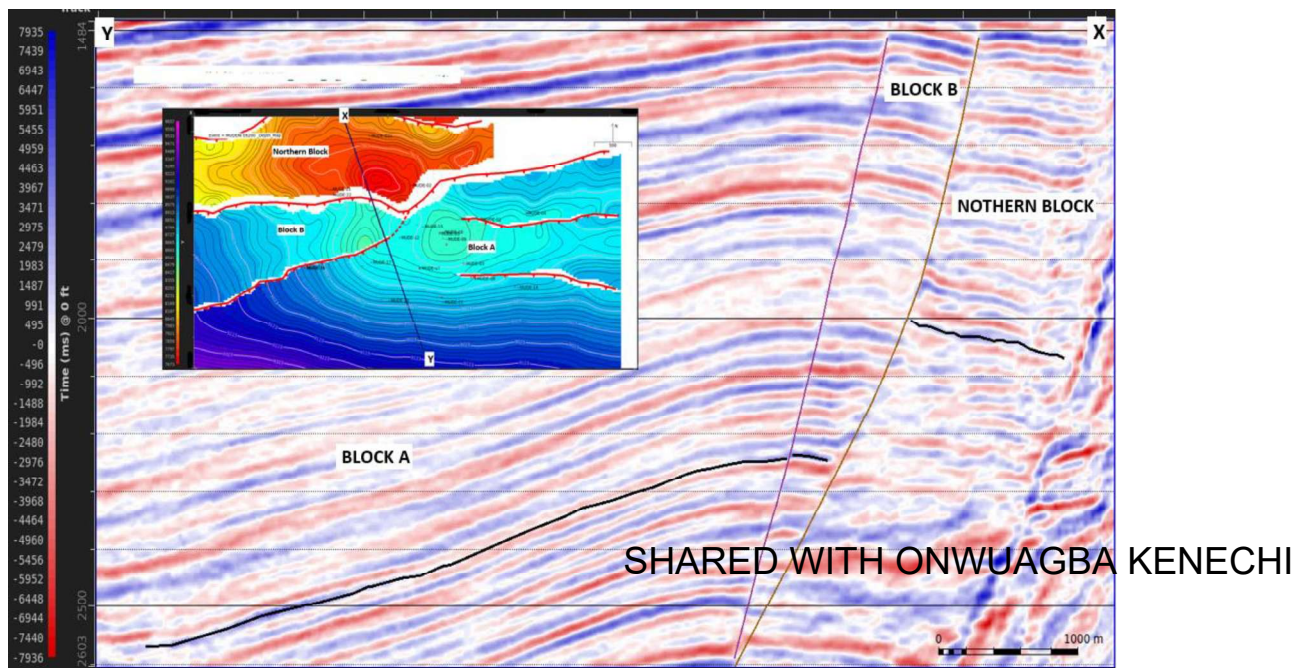


Figure 3: Mudeni E6200 Reservoir Blocks

CASE FOR STUDY

The Mudeni E6200 reservoir main block has been tested and proven by Mudeni well-01 in 1960. Production commenced in 1963 and to date, 23 wells have been drilled targeting the main block. The other block B is currently carried as an appraisal opportunity as it is yet to be penetrated by any well. Given the need to have a good estimate of potential volumes available in the untested

Mudeni E6200 block B, for appraisal and development planning purposes, the uncertainty around the fault sealing potentials of the minor fault between reservoir block 'A' and 'B' had to be adequately addressed. This is more pertinent because substantial volumes had been produced over the years from the main block. The implication of compartmentalization for field development strategy is that each block must be targeted separately by development wells. The Mudeni E6200 reservoir has produced the P1 proved reserves estimate and is still producing though the produced volume is still below the ultimate recovery estimates. A non-sealing fault would imply that significant volumes may have migrated across from block B into the zone of production in block A. Inter-reservoir block connectivity uncertainty was therefore a key issue that would impact on any further development strategy for the Mudeni Field (Fig. 4).

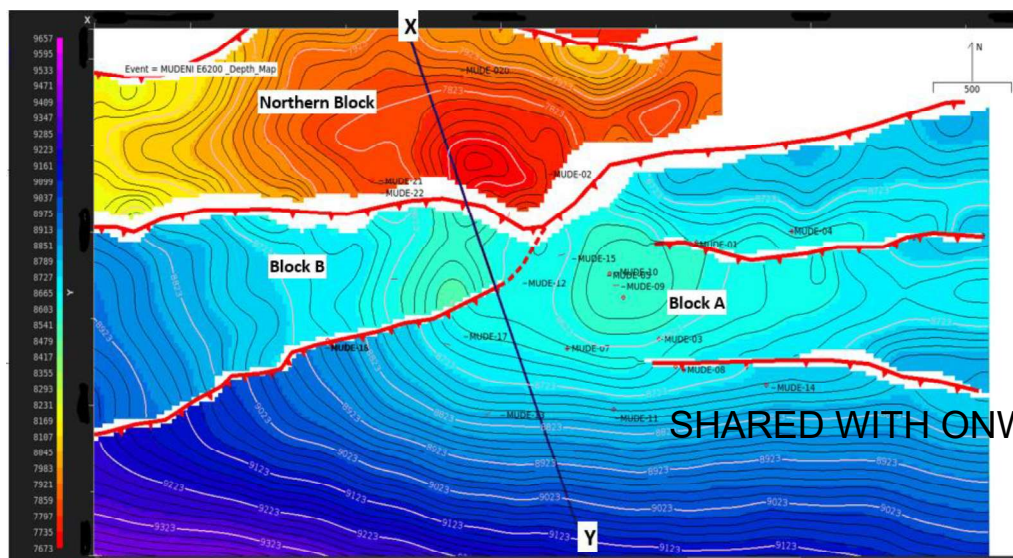


Figure 4: Mudeni E6200 Depth Map Showing Inter-Reservoir Connectivity Uncertainty between Block 'A' and Block 'B'.

DATA AND METHODOLOGY

Several literatures have documented the relationships between reservoir pore fill types and the corresponding acoustic response due to bulk density and seismic velocity properties (Prskalo S., 2004). The magnitude of reflected energy from a normally incident seismic waveform depends on the acoustic impedance contrast across the boundary of incidence (Equation 1). In a clastic reservoir with significant interconnected porosities and reasonable permeability, constituent fluid types are capable of segregation into clearly defined columns in response to the Earth's gravitational effects on the respective fluids due to their distinct density properties (Fig. 5). The fluid contact boundaries between the respective fluid columns could be identified based on their seismic response on a reflectivity data. Seismic response from a hydrocarbon bearing zone along the top of structure is generally distinguishable from the response of a brine filled zone (background value) in young clastic sediments with significant porosities. The initial model built for the Mudeni E6200 reservoir was that of a non-sealing fault with a common fluid contact of 8656ft tvdss as logged from wells in block 'A' being applied to both reservoir blocks (Fig. 6).

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\rho_2 \cdot v_2 - \rho_1 \cdot v_1}{\rho_2 \cdot v_2 + \rho_1 \cdot v_1} \quad \text{----- (1)}$$

Where:

R is the seismic reflectivity response from the interface

Z is the impedance property of the rock

ρ is the bulk density of formation rock

v is the seismic velocity of propagation.

The subscripts 1 and 2 represent the upper and lower formation rocks respectively.

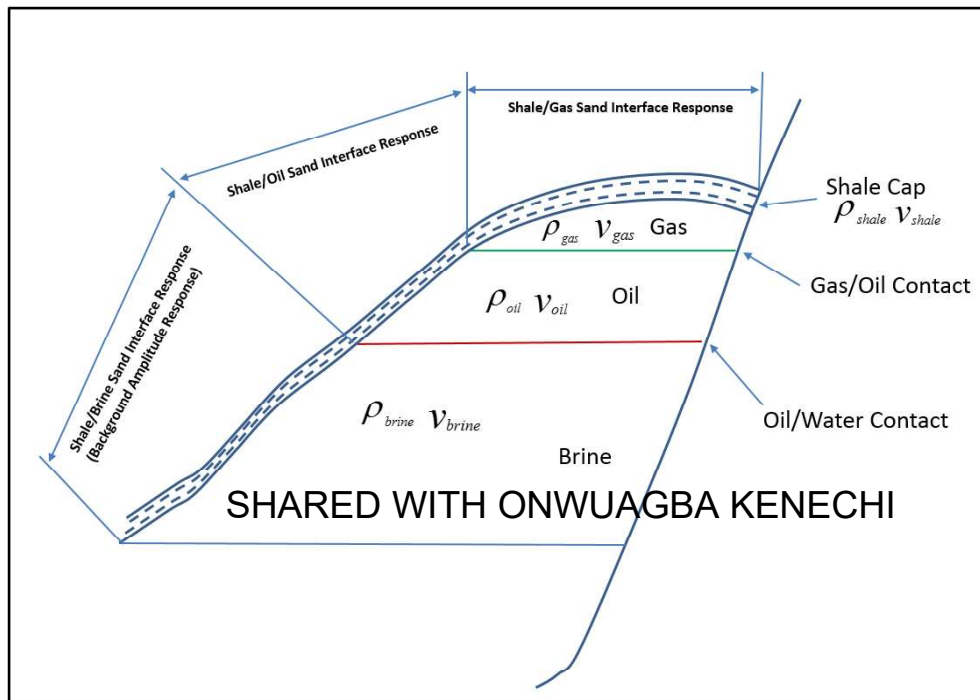


Figure 5: Segregation of Gas, Oil and Brine in a Reservoir in Response to Gravity. The Figure also shows zones from where seismic responses of the different fluid columns could be measured along the top structure.

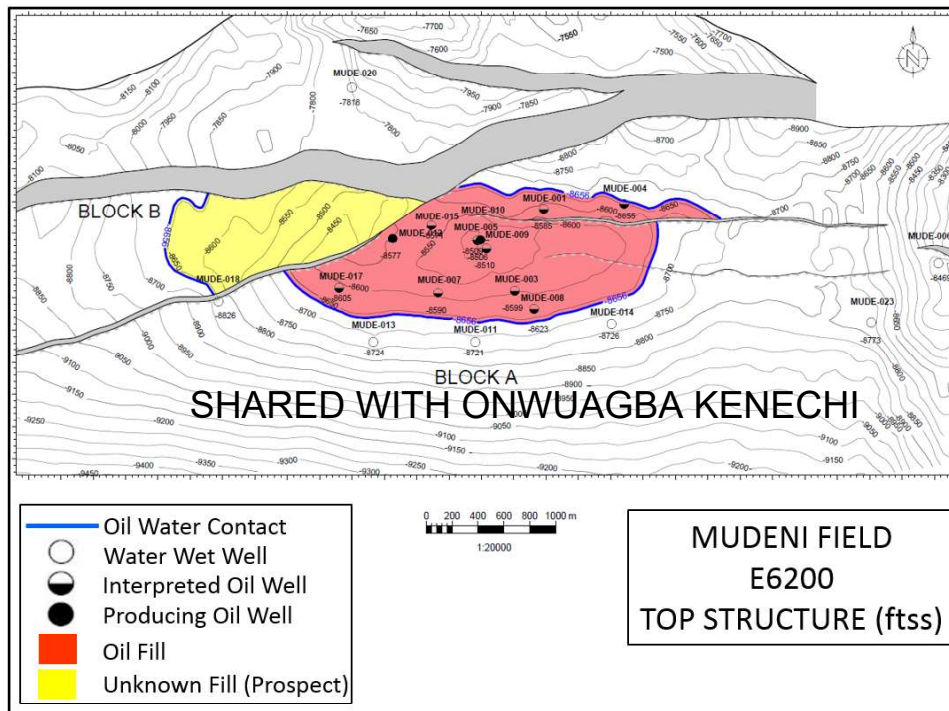


Figure 6: Mudeni Field Map based on non-sealing fault Scenario. With Common Fluid Contact of 8656 ft ss carried for Blocks A and B.

A detailed mapping of the E6200 seismic event on a 2014 PSDM seismic data provided the basis for guided seismic amplitude extraction. The extracted amplitudes from the E6200 reservoir was checked for conformability with the top structure. Using depth converted time grids, amplitude versus depth profiles were generated and analyzed for blocks A and B. Hydrocarbon–Water Contacts were estimated based on observed amplitude profiles. The fluid contact prediction methodology validation was done by comparing prediction results with actual data for Mudeni E6200 reservoir block A.

RESULTS AND DISCUSSION

The extracted seismic amplitudes for the Mudeni E6200 reservoir showed reasonable conformability with the reservoir top structure at blocks ‘A’ and ‘B’ respectively (Fig. 7). The observed amplitude versus depth profile for Mudeni E6200 reservoir block A showed essentially a single hump rising above the background response which is typical of a two fluid phase condition of hydrocarbon on brine (Fig. 8). The E6200 level is known to be oil bearing in Block ‘A’ based on results from wells drilled into the reservoir. The observed top structure seismic amplitude response value at block ‘A’ was therefore calibrated to oil response based on the well data. The Contact prediction technique validation was successful as it confirmed the logged hydrocarbon water contact in block ‘A’ at a depth of 8656ft tvdss as measured from existing wells (Fig. 8). The amplitude versus depth cross plot for Mudeni E6200 reservoir block B showed similar profile when compared with that from block ‘A’ and is indicative of a two fluid phase system of oil on brine. Predicted Hydrocarbon-Water Contact in block B was estimated to be at a depth of 8723 fttvd ss which is 67 ft deeper than the contact at block A (Fig. 9). The result of the study suggests that Block ‘A’ and Block ‘B’ are compartmentalized being found to have different fluid contacts.

An updated model for the Mudeni E6200 reservoir was built with the scenario for which reservoir blocks A and B are not in communication having two separate oil water contacts of 8656 ftss and 8723 ftss respectively (Fig. 10). The implication is that Mudeni E6200 reservoir block 'B' must be appraised and proven by a separate well for development purposes.

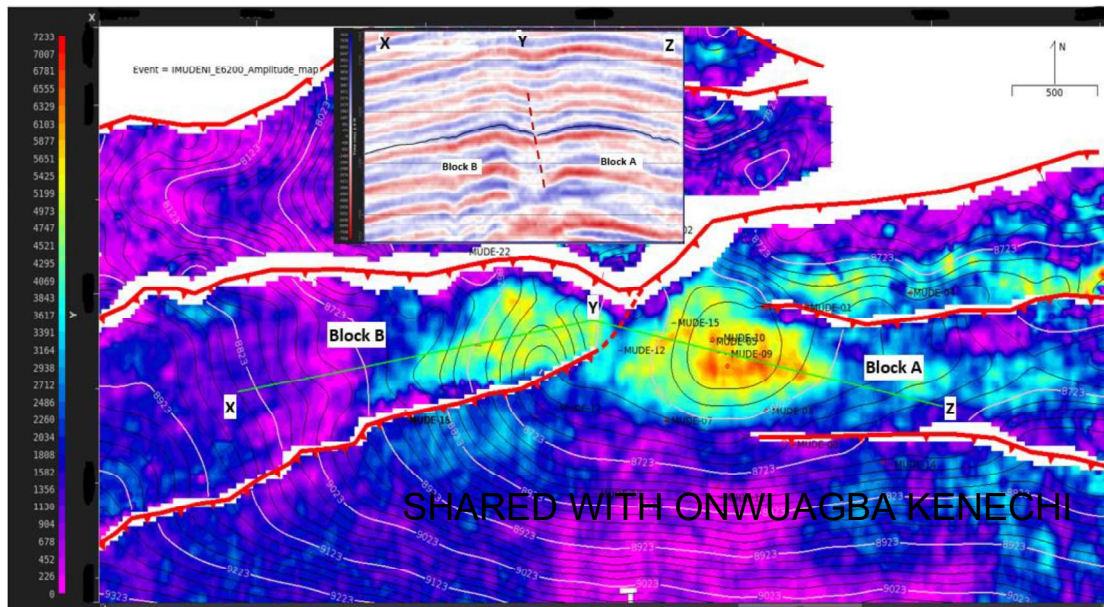


Figure 7: Mudeni E6200 Amplitude map showing conformity with top structure contours.

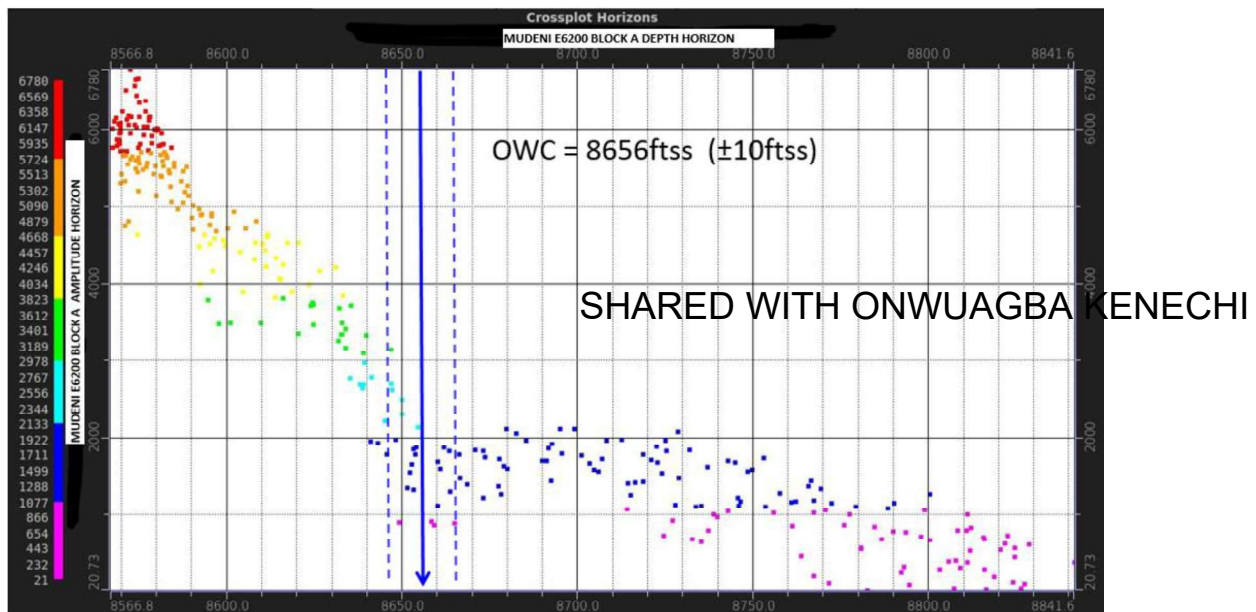


Figure 8: Mudeni E6200 Reservoir Block 'A' Amplitude Versus Depth Cross Plot.

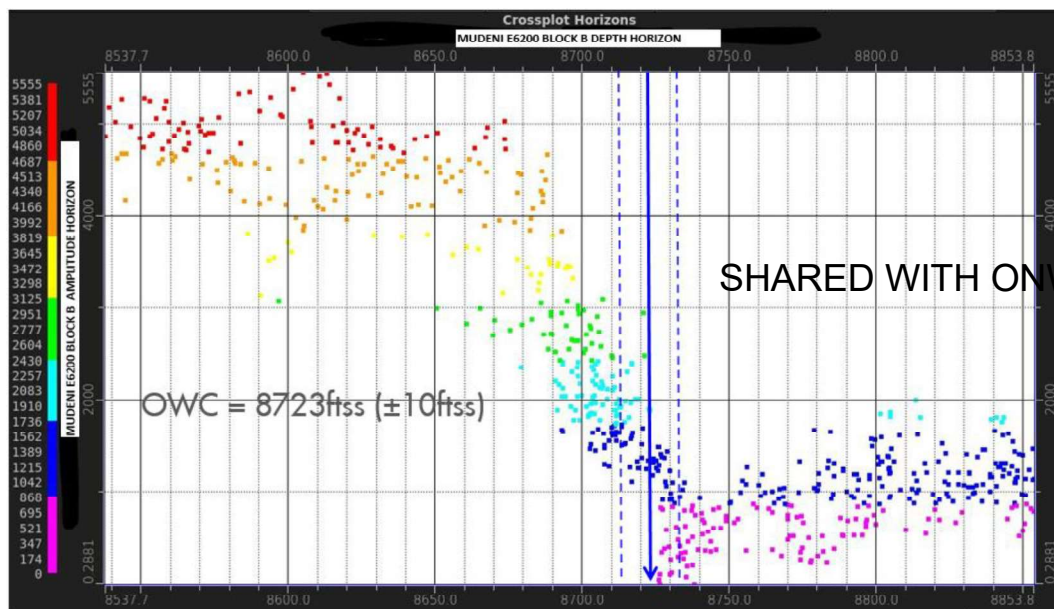


Figure 9: Mudeni E6200 Reservoir Block B Amplitude Versus Depth Cross Plot

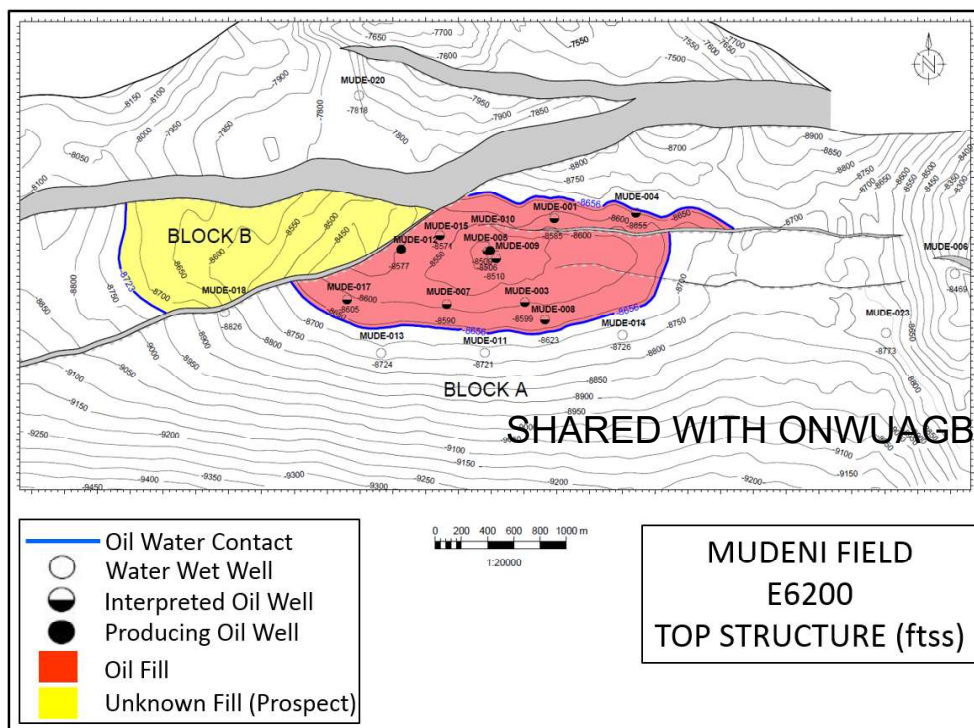


Figure 10: Mudeni E6200 Reservoir Map based on a Sealing Fault Scenario with Different Fluid Contacts of 8656 ftss and 8723 ftss carried for Block 'A' and 'B' respectively.

CONCLUSION

Uncertainty in Mudeni E6200 reservoir fault block connectivity was successfully resolved by using seismic amplitude attribute for investigation of the presence of hydrocarbon fill as well as estimation of fluid contacts in E6200 A and E6200 B reservoir blocks. Seismic amplitude conformability to reservoir top structure considered alongside with the amplitude vs depth profiles formed the technical bases for fluid contact prediction for each reservoir block.

A significant difference of 67 ft in predicted fluid contacts between Mudeni E6200 block A and block B reservoirs gave clear indication that the fault is sealing and that both reservoir blocks are compartmentalized. The implication of this is that the Mudeni E6200 reservoir block B should be appraised and proven with a separate well drilled to target that block as part of field development strategy.

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